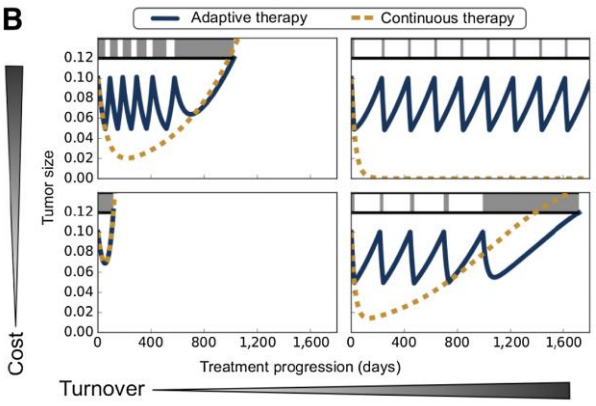
What Should Adaptive Cancer Therapy Optimize? A Reinforcement Learning Comparison

Team: Arnav Kumar (MD24B003) & Rohit Satish Chipde (MD24B026)

Clinical Problem: Standard cancer treatments often use the Maximum Tolerated Dose (MTD) to kill cancer cells as fast as possible. However, this approach kills off sensitive cells and leaves behind drug-resistant cells. Without competition from the sensitive cells, the resistant ones grow out of control. Adaptive therapy tries to fix this by stopping the drug when the tumor shrinks, keeping some sensitive cells alive to suppress the resistant ones.



Gallagher K, Strobl MAR, Park DS, et al. Mathematical Model-Driven Deep Learning Enables Personalized Adaptive Therapy. *Cancer Res*. 2024;84(11):1929-1941.

Objective / Research Problem: Researchers are using Reinforcement Learning (RL) AI to figure out the best adaptive therapy schedules. The problem is that different research papers train their AI with completely different goals. Our objective is to test these different reward goals in a single environment to see if changing the AI's objective actually changes its treatment strategy.

Why this Problem is important: If all goals lead to the same treatment plan, researchers can stop arguing about AI rewards. If they lead to different plans, doctors need to know exactly the hidden costs of trusting an AI trained on the 'wrong' objective.

Dataset Used: We are not using raw patient datasets. Instead, we are using a fitted two-population Lotka-Volterra ordinary differential equation (ODE) model from the paper by Strobl et al. (2021). The parameters in their ODE model were originally fitted using clinical data from 67 real prostate cancer patients.

Input and Output: Input: The current state of the tumor (total tumor size). Output: The action the AI chooses to take (give a full dose of the drug, a partial dose, or pause the drug).

Proposed Methodology: We will train several RL agents (using the PPO algorithm) in the Strobl ODE environment. We will train different agents using reward objectives taken from major papers: Survival (Gallagher 2023), Dose & Time (Gluzman 2020), Coexistence (Giles 2026), and Durability (Yao 2024). Once trained, we will perform a full cross-evaluation: testing every agent on every metric to see how they perform outside of what they were specifically trained for.

Baseline Models: To prove our AI works, we will compare it against standard, non-AI rules: No treatment, Continuous MTD, standard Adaptive Therapy (like AT50, which pauses at 50% tumor size), and 'Value Iteration' (a mathematical calculation of the absolute perfect score for each objective). We will also recreate the results given in Strobl paper to ensure there is no issue with the environment

Evaluation Metrics: Time to Progression (days until the tumor grows too big), total cumulative drug dose, time off the drug (holidays), and the final percentage of resistant cells.

Expected Outcomes: We expect to create a cross-evaluation matrix that proves one of two things: either all AI objectives converge on the same 'bang-bang' threshold policy (meaning the environment matters more than the reward), or they diverge significantly, showing the exact clinical cost of training an AI on the wrong objective.

Division of Work:

Arnav Kumar: Set up the ODE environment, implement the baseline models, and reproduce the initial paper parameters. He will also train the PPO agents.

Rohit Chipde: Figure out the different RL reward objectives, and generate the final cross-evaluation metrics and graphs.

References:

• Strobl et al. (2021). Cancer Res. 81(4):1135. • Gallagher et al. (2023/24). Cancer Res. • Giles & Newton (2026). Phys. Rev. E. • Yao et al. (2024). Brief. Bioinform.